19.4-Guided Exercise: Configure SSH Key-based Authentication

Ahmed Abdelhedi

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```
[operator1@servera ~]$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/operator1/.ssh/id_ed25519):
Created directory '/home/operator1/.ssh'.
Enter passphrase for "/home/operator1/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/operator1/.ssh/id_ed25519
Your public key has been saved in /home/operator1/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:0zTrRyPM8Um0jObPxigdtpYOJVm2xC24ROFurKQgq7w operator1@servera
The key's randomart image is:
+--[ED25519 256]--+
|        o.  .    |
|       o o = .   |
|        + % =    |
|       + & O .   |
|. .   . S % =    |
| o . o o B @ .   |
|.   . . o B *    |
|o        = o     |
|.E.       .      |
+----[SHA256]-----+
[operator1@servera ~]$
```

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[operator1@servera ~]$ ssh-copy-id operator1@serverb
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/operator1/.ssh/id_ed25519.pub"
The authenticity of host 'serverb (172.25.250.11)' can't be established.
ED25519 key fingerprint is SHA256:iS9Q0/bXHOwXKX5zTP10nAIxSnKQoUQxMfKE0s0b2ZU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
operator1@serverb's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'operator1@serverb'"
and check to make sure that only the key(s) you wanted were added.

[operator1@servera ~]$
```

```
[operator1@servera ~]$ ssh operator1@serverb
Enter passphrase for key '/home/operator1/.ssh/id_ed25519':
Register this system with Red Hat Insights: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
[operator1@serverb ~]$
```

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```
[operator1@servera ~]$ ssh-keygen -f ~/.ssh/protected-key
Generating public/private ed25519 key pair.
Enter passphrase for "/home/operator1/.ssh/protected-key" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/operator1/.ssh/protected-key
Your public key has been saved in /home/operator1/.ssh/protected-key.pub
The key fingerprint is:
SHA256:KMsRDmrHhtudHEsmCUjNRgWdY5nQmm/PXowp+L0vzjw operator1@servera
The key's randomart image is:
+--[ED25519 256]--+
| .++*.+          |
|o  + O           |
|..o = .          |
| .+=.. .         |
|.o *=+. S        |
|. =.B*+ +        |
| . o+*oo o       |
|    . =E.        |
|     .oB=.       |
+----[SHA256]-----+
[operator1@servera ~]$
```

```
[operator1@servera ~]$ ssh-copy-id -i ~/.ssh/protected-key.pub operator1@serverb
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/operator1/.ssh/protected-key.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
Enter passphrase for key '/home/operator1/.ssh/id_ed25519':
Enter passphrase for key '/home/operator1/.ssh/id_ed25519':
operator1@serverb's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh -i /home/operator1/.ssh/protected-key 'operator1@serverb'"
and check to make sure that only the key(s) you wanted were added.

[operator1@servera ~]$
```

```
[operator1@servera ~]$ ssh -i ~/.ssh/protected-key operator1@serverb
Enter passphrase for key '/home/operator1/.ssh/protected-key':
Register this system with Red Hat Insights: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
Last login: Mon Sep 28 20:04:38 2026 from 172.25.250.10
[operator1@serverb ~]$
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```
[operator1@servera ~]$ eval $(ssh-agent)
Agent pid 3867
[operator1@servera ~]$ ssh-add ~/.ssh/protected-key
Enter passphrase for /home/operator1/.ssh/protected-key:
Identity added: /home/operator1/.ssh/protected-key (operator1@servera)
[operator1@servera ~]$ ssh -i ~/.ssh/protected-key operator1@serverb hostname
serverb
[operator1@servera ~]$
```

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```
[operator1@servera ~]$ exit
logout
[student@servera ~]$ exit
logout
Connection to servera closed.
student@workstation:~$ lab finish ssh-keyauth
🤖 Running: lab finish ssh-keyauth
SUCCESS Remove operator1 user in servera
SUCCESS Remove operator1 user in serverb
student@workstation:~$
```